

Basic Setup:

$$y_{n \times 1} = X_{n \times p} \beta_{p \times 1} + \epsilon_{n \times 1}$$

$$\hat{y}_{n \times 1} = H_{n \times n} y_{n \times 1}$$

$$H = X(X^T X)^{-1} X^T$$

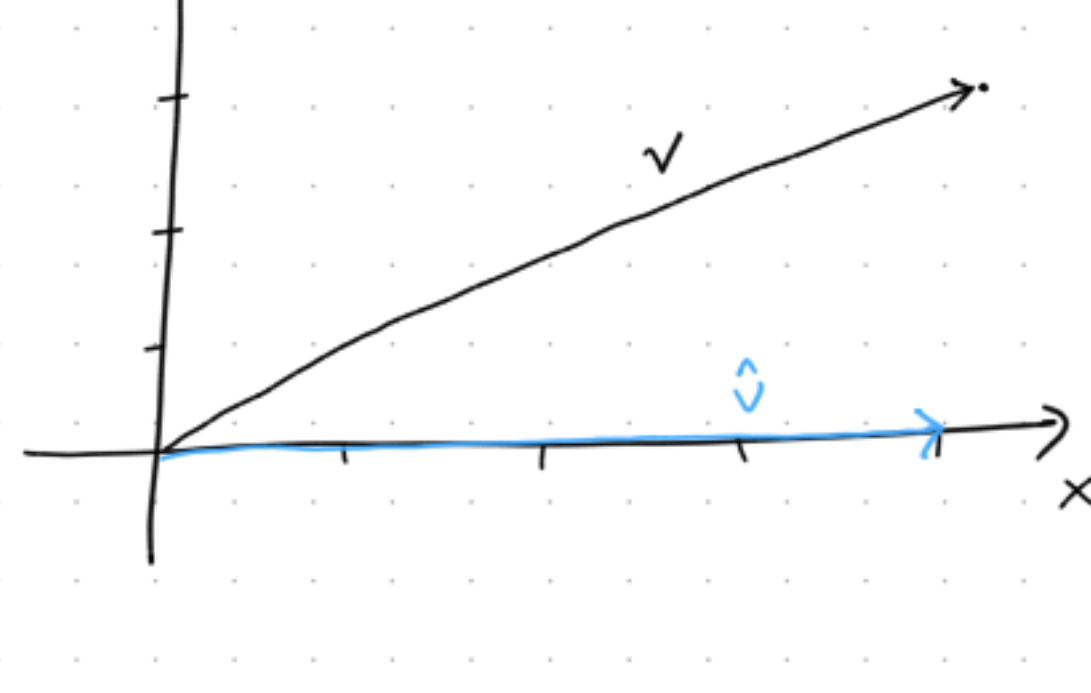
(Claim: H is a projection matrix

1) H is symmetric $\Rightarrow H^T = H$

2) H is idempotent $\Rightarrow H^2 = H$

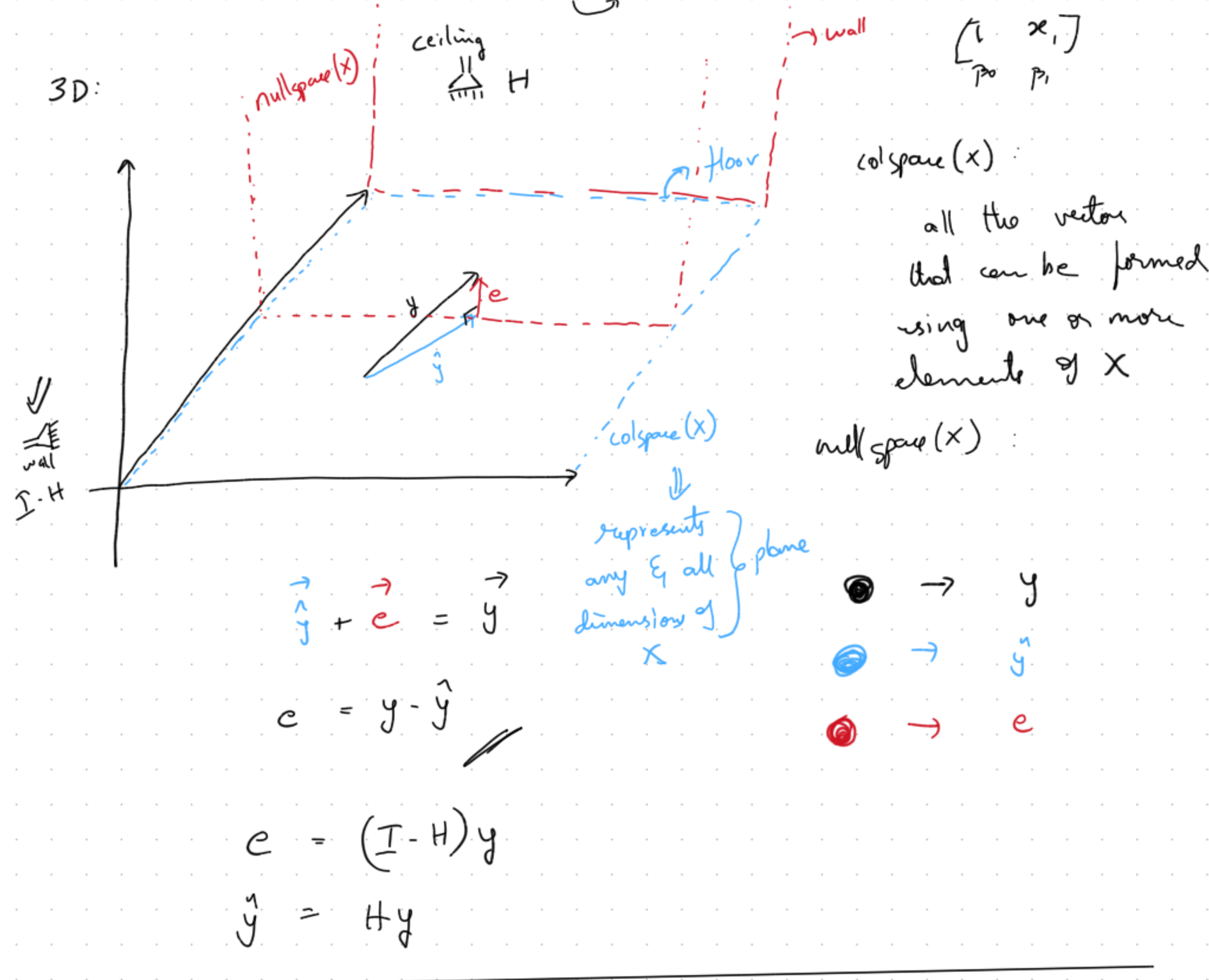
light source

$$\vec{v} = 4\hat{i} + 3\hat{j}$$



$$O_{n \times p} = \vec{0}$$

The amount of pattern / variance / prediction that can be explained / guessed / estimated using our models / analysis is basically $\hat{y}$ $X \rightarrow$ fixed



MLR 12 revised

find unbiased estimator of σ^2

$$e \sim N(0, (I - H)\sigma^2)$$

$$\frac{e}{(I - H)^{1/2}} \sim N(0, 1)$$

$$\frac{1}{\sigma} (I - H)^{1/2} (I - H)y \sim N(0, 1)$$

$$\frac{1}{\sigma} (I - H)^{1/2} y \sim N(b, 1)$$

$$Z^T Z = z_1^2 + z_2^2 + \dots + z_n^2 \sim \chi_{df}^2$$

$$Z^T Z = \left[\frac{1}{\sigma} (I - H)^{1/2} y \right]^T \left[\frac{1}{\sigma} (I - H)^{1/2} y \right] \sim \chi_{df}^2$$

$$= \frac{1}{\sigma^2} y^T (I - H)^{1/2} (I - H)^{1/2} y \sim \chi_{df}^2$$

$$= \frac{1}{\sigma^2} y^T (I - H) y \sim \chi_{df}^2$$

$$= \frac{1}{\sigma^2} y^T (I - H) y \sim \chi_{df}^2$$

$$= \frac{1}{\sigma^2} e^T e \sim \chi_{df}^2$$

$$\text{where } df = \text{rank}(I - H)$$

$$= \text{tr}(I - H)$$

$$\Rightarrow \text{tr}(I - H) = \text{tr}(I) - \text{tr}(H)$$

$$= \text{tr}(I_n) - \text{tr}(X(X^T X)^{-1} X^T)$$

$$= \text{tr}(I_n) - \text{tr}(X^T X (X^T X)^{-1})$$

$$= \text{tr}(I_n) - \text{tr}(I_p)$$

$$= n - p$$

$$\Rightarrow \frac{e^T e}{\sigma^2} \sim \chi_{n-p}^2$$

$$E\left[\frac{e^T e}{\sigma^2}\right] = n - p$$

(from def. of χ^2 below)

$$E\left[\frac{e^T e}{n - p}\right] = \sigma^2$$

$$\Rightarrow E[\hat{\sigma}^2] = \sigma^2 \quad (\text{from definition of unbiased})$$

$$\Rightarrow \hat{\sigma}^2 = \frac{e^T e}{n - p} \quad \text{unbiased estimator of } \sigma^2$$

$$\text{Var}(ax) = a^2 \text{Var}(x)$$

$$\text{if } \text{Var}(x) = K$$

$$\frac{\text{Var}(x)}{a} = \frac{K}{a}$$

$$\text{Var}\left(\frac{x}{a}\right) = \frac{K}{a}$$

$$X = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

$$X^T X = \sum x_1^2 + x_2^2 + \dots + x_n^2$$

$$I - H^T = I - H$$

$$(I - H^T)^{1/2} = (I - H)^{1/2}$$

$$(I - H)^{1/2} = (I - H)^{1/2}$$

$$e^T e = \left[(I - H)y \right]^T (I - H)y$$

$$= y^T (I - H)^T (I - H)y$$

$$= y^T (I - H)(I - H)y$$

$$= y^T (I - H)y$$

$$\text{tr}(ABC) = \text{tr}(CAB)$$

$$= \text{tr}(BCA)$$

$$I_n = \begin{bmatrix} 1 & 0 & 0 & \dots & 0 \\ 0 & 1 & 0 & \dots & 0 \\ 0 & 0 & 1 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & 1 \end{bmatrix}_{n \times n}$$

$$\text{tr}(I_n) = n$$

Side note:

1) Std. Normal:

$$z_i \sim N(0, 1)$$

2) Chi-sq distribution

$$V \sim \chi_{df}^2$$

$$\text{where } V = z_1^2 + z_2^2 + \dots + z_n^2$$

$$\text{where } z_i \sim N(0, 1)$$

$$E[V] = n - p, \quad \text{where } n \rightarrow \text{no. of obs.}$$

$$p \rightarrow \text{no. of parameters}$$

3) F distribution:

$$F_{df_1, df_2} \sim \frac{\chi_{df_1}^2 / df_1}{\chi_{df_2}^2 / df_2}$$

MLR - 8 study guide

$$CI \text{ for } y \Rightarrow \hat{y} \pm SE(\hat{y}) \times t_{\alpha/2, p}^*$$

$$SE(\hat{y}) = x_h^T SE(\hat{\beta}) x_h$$

$$\text{Var } \hat{\beta} = (X^T X)^{-1} \sigma^2$$

$$SE(\hat{\beta}) = \sqrt{\text{diag}((X^T X)^{-1} \sigma^2)}$$

$$CI \text{ for } \theta \Rightarrow \hat{\theta} \pm SE(\hat{\theta}) \times t_{\alpha/2, df}^*$$

$$y = \hat{y} \pm SE(\hat{y}) \times t_{\alpha/2, df}^* \Rightarrow \hat{y} \pm SE(\hat{y}) \times t_{\alpha/2, p}^*$$

$$\text{where, } df = \text{rank}(H)$$

$$= \text{tr}(H)$$

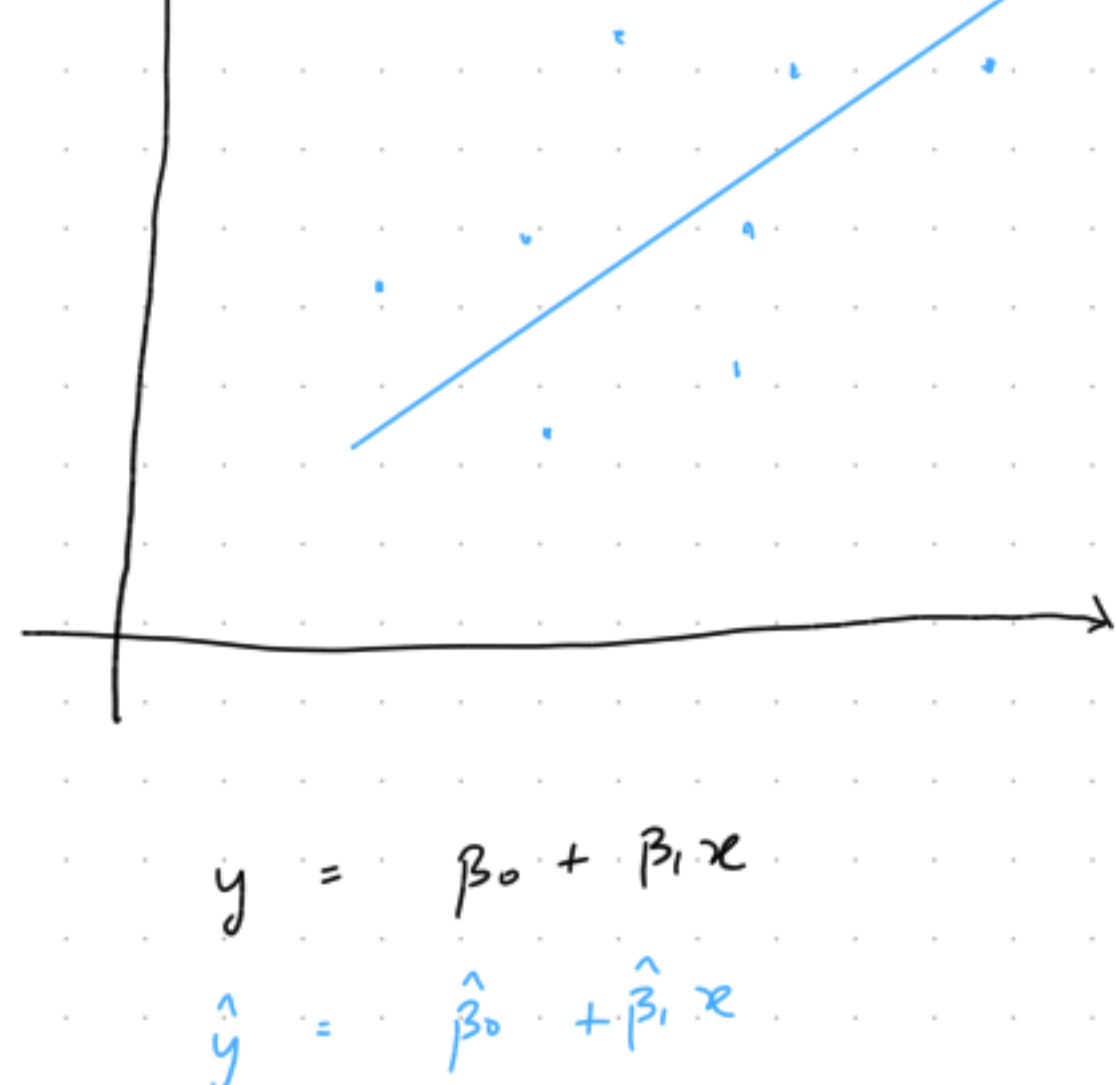
$$= \text{tr}(X(X^T X)^{-1} X^T) = \text{tr}\left(X^T X (X^T X)^{-1}\right)$$

$$= \text{tr}(I_p)$$

$$= p$$

Influential Point:

Conf:

determine if $\bullet$ is an influential point

$$y = \beta_0 + \beta_1 x$$

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$$

to determine if a point is influential or not we can see Cook's distance.

$$D_i = \frac{\sum (y_i - \hat{y}_i)^2}{p \hat{\sigma}^2} \times \frac{h_{ii}}{(1 - h_{ii})^2}$$

$$h_{ii} / x_i = 0 = \frac{1}{n} + \frac{(0 - \bar{x})^2}{SSX} = \frac{1}{n} + \frac{\bar{x}^2}{SSX}$$

$$D_{x=0} = \frac{\sum (y_i - \hat{\beta}_0)^2}{2 \hat{\sigma}^2} \times \left(\frac{\frac{1}{n} + \frac{\bar{x}^2}{SSX}}{(1 - \frac{1}{n} - \frac{\bar{x}^2}{SSX})^2} \right)$$

if $D > \frac{4}{n}$ then influentialSo just by knowing $x=0$, we can't say if it is influential or not.- If $\bar{x}$ is large, i.e. $x=0$ is far away from x_i 's then $x=0$ could be influential- If SSX is large i.e. predictors are widely spread then D could go down and $x=0$ may not be an influential point.- Similarly if e is large or small $\hat{\sigma}^2$ is large or small you can take corresponding decisions.

for SLR,

$$h_{ii} = H_{ii}$$

$$= X(X^T X)^{-1} X^T$$

$$X^T X = \begin{bmatrix} n & \sum x_i \\ \sum x_i & \sum x_i^2 \end{bmatrix}$$

$$(X^T X)^{-1} = \frac{1}{n \sum x_i^2 - (\sum x_i)^2} \begin{bmatrix} \sum x_i^2 & -\sum x_i \\ -\sum x_i & n \end{bmatrix}$$

$$e = \begin{bmatrix} 1 \\ x_i \end{bmatrix}$$

$$h_{ii} = x_i^T (X^T X)^{-1} x_i$$

$$= (1 \ x_i) \begin{bmatrix} \sum x_i^2 & -\sum x_i \\ -\sum x_i & n \end{bmatrix} \begin{bmatrix} 1 \\ x_i \end{bmatrix}$$

$$h_{ii} = \frac{1}{n} + \frac{(x_i - \bar{x})^2}{SSX}$$